Standard Operating Procedure (SOP) for Cisco PE Role HSR-PE-07: Mitigating Major Optical Transport Failure (ALARM_2522) at HSR Layout Access Site

Node: HSR-PE-07 | **Vendor:** Cisco | **Severity:** Major | **Alarm Code:** ALARM_2522

Description

The occurrence of a Major Optical Transport Failure (ALARM_2522) on Cisco PE Role HSR-PE-07 is attributed to the degradation of SFP/XFP transceivers, fiber cuts, or a decrease in received/transmitted power. This failure domain can be attributed to the inherent characteristics of optical transport systems, where signal attenuation and distortion can occur due to the physical properties of light transmission over long distances. The use of SFP/XFP transceivers on Cisco PE equipment allows for the seamless integration of optical signals, thereby enabling high-speed data transmission over the network. However, the reliability of these transceivers is critical to the overall network performance. In this scenario, the degradation of SFP/XFP transceivers or fiber cuts can lead to a significant decrease in received power, resulting in a Major Optical Transport Failure. Furthermore, the complexity of the Cisco PE equipment's control plane, which utilizes protocols such as OSPF and ISIS for routing and signaling, can also contribute to the occurrence of this failure domain. The ability to accurately diagnose and resolve this issue is crucial to prevent network downtime and ensure the continued delivery of high-quality services.

Pre-requisite Checks

- Pre-requisite: Verify site access and ensure all necessary safety precautions have been taken by dispatching remote hands to HSR Layout Access., Description: This will enable the engineer to safely access the site and perform the necessary diagnostic and resolution steps.

- Pre-requisite: Check the status of redundant links to ensure that they are functioning correctly., Description: This will allow the engineer to confirm that the network can continue to operate in the event of a failure on the primary link.

Potential Root Causes

Cause	Probability
Degradation of SFP/XFP transceivers.	High
Fiber cuts.	Medium
Decrease in received/transmitted power.	Low

Diagnostic Commands & Expected Outputs

```
Command: show int status, Expected Output: The interface is up and
operational., Description: This command will provide information on
the current status of the interface, including whether it is up and
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operational. This will help to determine if the issue is related to
the interface itself or the SFP/XFP transceiver.

Command: show interface transceiver detail, Expected Output: The
transceiver is operational and within the acceptable temperature
range., Description: This command will provide detailed information on
the SFP/XFP transceiver, including its operational status and
temperature. This will help to determine if the issue is related to
the transceiver itself.

Command: show interface stats, Expected Output: The interface
statistics are within the acceptable range., Description: This command
will provide information on the interface statistics, including the
number of packets transmitted and received. This will help to
determine if the issue is related to the interface itself or the
SFP/XFP transceiver.
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Resolution Steps

Step 1: Step: Replace the SFP/XFP transceiver., Command: interface , Description: This command will enable the engineer to access the specific interface and replace the SFP/XFP transceiver. The new transceiver should be installed and the interface should be re-enabled to ensure proper operation.

Step 2: Step: Verify the fiber connection., Command: show interface transceiver detail, Description: This command will provide detailed information on the SFP/XFP transceiver, including its operational status and temperature. The engineer should verify that the fiber connection is secure and that the transceiver is within the acceptable temperature range.

Step 3: Step: Adjust the received/transmitted power., Command: controller power-adjust, Description: This command will enable the engineer to adjust the received/transmitted power on the interface. The engineer should verify that the power levels are within the acceptable range to ensure proper operation.

Post-Check Verification

```
Command: show interface status, Expected Output: The interface is up
and operational., Description: This command will provide information
on the current status of the interface, including whether it is up and
operational. The engineer should verify that the interface is
operational and that the SFP/XFP transceiver is functioning correctly.

Command: show interface transceiver detail, Expected Output: The
transceiver is operational and within the acceptable temperature
range., Description: This command will provide detailed information on
the SFP/XFP transceiver, including its operational status and
temperature. The engineer should verify that the transceiver is
operational and within the acceptable temperature range.
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Roll-back Procedures

- Step: Restore the original SFP/XFP transceiver., Command: interface , Description: This command will enable the engineer to restore the original SFP/XFP transceiver and re-enable the interface to ensure proper operation.

- Step: Verify the fiber connection., Command: show interface transceiver detail, Description: This command will provide detailed information on the SFP/XFP transceiver, including its operational status and temperature. The engineer should verify that the fiber connection is secure and that the transceiver is within the acceptable temperature range.